THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE CHAPTER TEST SET B

Chapter 2: **Acids, Bases and Salts** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following is an olfactory indicator? 1

(a) onion **(b)** litmus **(c)** turmeric **(d)** phenolphthalein

2. Phenolphthalein in a basic solution is: 1

(a) yellow **(b)** colourless **(c)** red **(d)** pink

3. A metal oxide reacts with hydrochloric acid to form salt and water. The metal oxide is: 1

(a) amphoteric only **(b)** basic **(c)** neutral **(d)** acidic

4. A student states: “Acids react with all metals to give hydrogen gas.” The correct response is: 1

(a) Correct
(b) Incorrect — less reactive metals like copper do not give hydrogen
(c) Correct for dilute acids
(d) Cannot be determined

5. Student A says the pH of milk becomes lower when it turns into curd; Student B says it becomes higher. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

6. The chemical name of bleaching powder is: 1

(a) sodium hypochlorite **(b)** calcium oxychloride
(c) calcium chloride **(d)** calcium carbonate

7. Which acid is present in tomato? 1

(a) citric acid **(b)** tartaric acid **(c)** oxalic acid **(d)** lactic acid

8. The pH of a solution is 2. It is: 1

(a) basic **(b)** strongly acidic **(c)** weakly acidic **(d)** neutral

9. The process of dissolving an acid in water is: 1

(a) endothermic **(b)** a decomposition
(c) neither **(d)** highly exothermic

10. The salt formed from a strong acid and a strong base has pH: 1

(a) less than 7 **(b)** more than 7 **(c)** 7 **(d)** 0

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. While diluting an acid, why is it recommended that the acid be added to water and not water to the acid? 2

12. Write the formula and one use of washing soda. 2

13. Why does an aqueous solution of an acid conduct electricity? 2

14. A student says that alcohol and glucose solutions are acidic because they contain hydrogen. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Name the gas evolved and write the equation when 3
(i) zinc reacts with dilute sulphuric acid
(ii) sodium hydrogencarbonate reacts with dilute HCl
(iii) zinc reacts with sodium hydroxide.

16. What is baking powder? How does it make cakes soft and spongy? Why does a cake made with only baking soda taste bitter? 3

17. What is the pH scale? How is pH related to the concentration of H^+ ions? Why does a farmer add quicklime to his field? 3

OR

Write two uses each of bleaching powder and baking soda. 3

18. Fresh milk has pH 6. How will the pH change as it turns into curd? Why does a milkman add a very small amount of baking soda to fresh milk? 3

OR

Explain how acidic and basic solutions behave with 3
(i) blue litmus
(ii) red litmus
(iii) methyl orange.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) What is the common name of the compound $CaOCl_2$? 5
(ii) How is it manufactured? Write the equation.
(iii) Give three uses.

OR

(i) Describe an activity to show that acids produce hydrogen gas when they react with metals. 5
(ii) Write the equation for the reaction of zinc with dilute HCl.
(iii) How is the gas tested?

20. **Case study:** The pH of the mouth falls after eating sweets, leading to tooth decay. 5
(i) Below what pH does tooth enamel corrode?
(ii) Which substance forms the tooth enamel?
(iii) How do bacteria lower the pH?
(iv) How does toothpaste help?
(v) Is toothpaste acidic or basic?

OR

(i) What is the chlor-alkali process? Why is it so called? 5
(ii) Write the equation.
(iii) Give one use each of the three products.

END OF QUESTION PAPER — Total: 40 Marks